CSA
Cybersecurity Industry Call for Innovation
Submission Template

Project Title:	**Smart Home Security App**
Full Company Name:	**Ambient Systems Pte. Ltd.**

Notes to Applicant

1. Follow the slide template sequence strictly, i.e. do not change it
2. Each proposal should only respond to **ONE** Challenge Statement
3. All questions must be answered and preserved, i.e. do not delete any questions
4. Do not exceed 20 slides (excluding title slide, instructions/guidelines, separators, Thank You slide)
5. Convert the slides to pdf before submission
6. Maximum file size for submission: 7MB

CSA CCDF Guidelines

Extract of CSA Cybersecurity Co-Innovation and Development Fund (CCDF) guidelines

- No retrospective applications will be accepted
  - An application is deemed retrospective if the proposed project has already commenced at the time of application
- There must be development of new features, not just integration/configuration of existing products/solutions
- At least 50% of manpower requested for funding must be Singaporean or Singapore PR
- All Singapore registered companies are eligible for the CSA CCDF
  - Overseas firms that are not registered in Singapore will need to partner with a Singapore registered company
- Funding awarded must be used to carry out development activities physically in Singapore
  - Recipients of CCDF funding should use Singapore as a base to own, manage and register all intellectual property developed during the project
- Proposals already funded or considered for funding by other government agencies will not be considered for CCDF
- All claims will be disbursed on a reimbursement basis upon submission of claim document(s) such as salary slips, invoices etc.

Please refer to https://www.csa.gov.sg/programmes/co-innovation-development-fund for the full guidelines

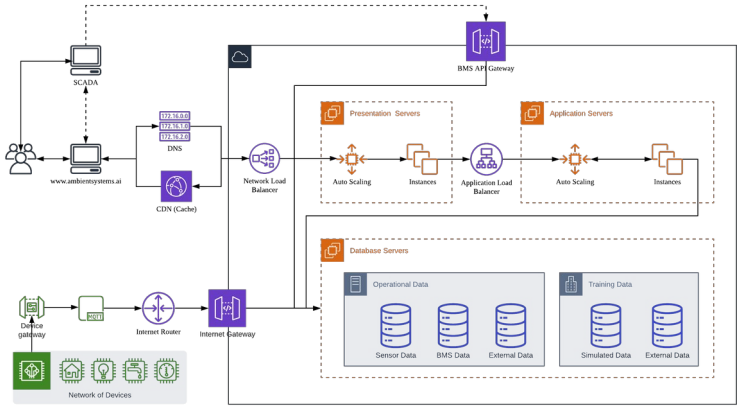
Project Title: Smart Home Security App
Company Name: Ambient Systems Pte. Ltd.

Project Quad Chart

Summary

Technical Lead: and CEO/[redacted]

Illustration



Objectives / Deliverables

Challenge Statement

- State the Challenge Statement proposal addresses
  - ❑ CS07: Unified IoT Security for Connected Products Utilising

Objective

- Provide an application that securely manages intelligent devices in a home network

Deliverables

- Fully functioning application

TRL (Technology Readiness Level)

- Start TRL = <3>; Projected End TRL = <9>

Approach and Innovation*

Proposed Approach

➔ Leverage device sensor data and classification models (using machine learning) to categorize approved device behaviour

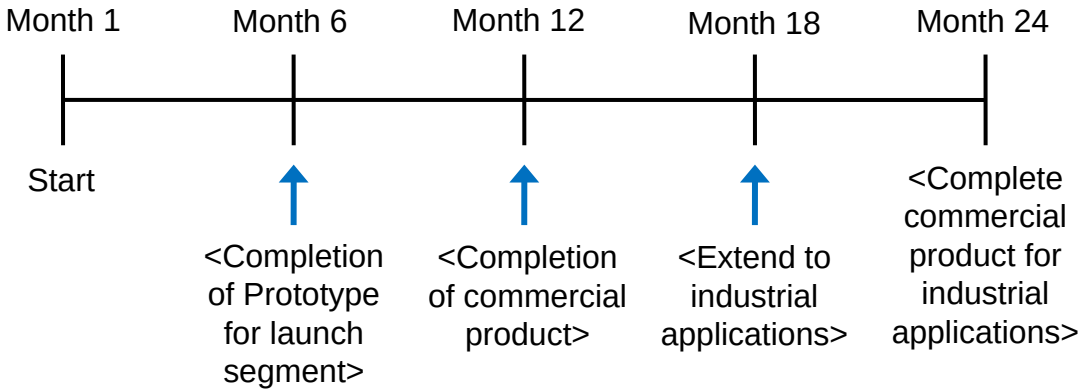
Innovation

- The application leverages network metadata and device data to evaluate the risk profile of the devices
- Flags anomalous device behaviour for investigation or remediation

*__This section is mandatory and will be important for the evaluation__

Funding and Schedule

- **ed**: **<S$ 540,000>**
- **Total Project Cost: <S$ 540,000>**
- **Schedule (Max 24 months)**: 4 significant project milestones



1. Objective and Approach

a) What problem is this proposal trying to solve?

Improving cybersecurity of edge devices in smart homes. In recent years, there has been an increase in the sources of input that may not be trusted. Furthermore, huge number of devices in homes may be difficult to manage. Some critical IoT devices also contribute to health and safety of the occupants, such as in the case of life-supporting or home security devices in homes. The end result is an increased attack surface for intruders to gain control of user's network, device, or personal data.

b) What does the proposed solution do? How will it solve the problem?

The proposed app pairs the IoT devices and performs key functions to enhance overall cybersecurity. The pairing process leverages established protocols on Wi-Fi or Bluetooth and augments them using more advanced techniques such as context-based pairing. The app performs fingerprinting of device behaviour using Machine Learning classification algorithms, such as K-Means clustering. Each feature of the device will correspond to one cluster. These will be maintained between the app and the device. After the baseline is established, either in a lab setting or real-world deployment, the app will monitor the network and device for anomalies. Anomalies will be flagged for remediation by the user (where possible) and relayed to the control centre for evaluation. This approach not only enhances cybersecurity of the system but monitors the system for signs of hardware malfunctioning or tempering.

1. Objective and Approach

c) Challenge Statement Requirements: State which requirements the proposal will fulfil and which it will not

Delete this slide if responding to CSOC (Open Category)

No.	Requirement	Complied
1	Integrated approach in the detection of security configuration	Yes
2	Monitoring of end-to-end connections between connected devices and the entire hub	Yes
3	Have /federated learning abilities at the connected device	Yes
4	Automatically detect and alert a central controller	Yes
5	Provide telemetry in industry standard formats	Yes
6	Identify compromised devices and anomalous network activities based on an AI/ML engine	Yes
7	Detect and tag personal identifiable information	Yes
8	Attest to independent accreditation of applicable IoT security standards	Yes
9	Developed in a cloud-first approach	Yes
10	Shared library with open API approach embedded into the connected devices	Yes
11	Not adversely impact the system and network performance and operations	Yes

2. Proposed Solution Implementation

a) How does the proposed solution work? How will it be implemented?

The application can be deployed on a specialized IoT gateway or through the user's mobile device. Using the user's existing device would remove the need for new hardware, minimizing the cost of deployment. Furthermore, running the app on user's mobile device would not require modification to the existing IoT products. Most mobile devices already have extensive capabilities when it comes to sensing and networking. For instance, iPhone's existing pairing protocols on Wi-Fi and Bluetooth can be used with minimal modifications.

When the connected devices are paired with the mobile device (or a dedicated IoT gateway), the app will be able to perform its key functions. These include localization, pre-processing, classification, encryption, and cloud communications. Some of these functions will be leveraging ML models base on a rule book from historical device data. The ML models can be deployed on the Cloud or at the Edge, depending on computing capabilities of the mobile device or the IoT Gateway. The initial training will have to be done on the cloud and may leverage anonymized data from similar devices.

2. Proposed Solution Implementation

b) What are the project outputs or upon project completion?

No.	and Specifications
1	App securely pairs select devices
2	App monitors network traffic by the device
3	App monitors device vitals
4	App sends device vitals to the cloud
5	ML model (on the cloud) fingerprints device features
6	ML model (at the edge) monitors device features
7	App detects anomalous behavior (e.g., MitM attack)
8	App detects hardware malfunctioning (e.g., damaged sensors)
9	App securely pairs select devices

3. Innovation

a) What is the innovation exhibited by the proposed solution?

There are two innovative aspects of the solution. The first aspect is how the application is deployed. It achieves interoperability by pairing with the devices at the local network. The second innovative aspect is the approach to device fingerprinting, which leverages machine learning classification algorithms. The app uses clustering methods to group device behaviour into feature sets, creating a baseline for normal device behaviour. Anomalous behaviour is flagged for remediation.

b) Market Comparison on the Proposed Product with Other Products in the Market

No	Features	Your Product	LG ThinQ	Google Home
1	Open to multiple device vendors	Y	N	Y
2	Cybersecurity features	Y	N	N
3	Doesn’t required new or specialized hardware for the user	Y	N	N
4	AI / ML capabilities	Y	Y	Y
5	AI / ML capabilities for cybersecurity	Y	N	N

4. User Acceptance Test

a) What is the test setup? (include diagram to explain and list any assumptions)

- We plan to work with the project sponsor to demonstrate the application
- The app simply pairs with existing devices and leverages available sensor data and equipment metadata to enhance overall security of the devices and the network
- The profiling feature of the application must achieve a pre-determined level of accuracy in order to be accepted in a commercial application

4. User Acceptance Test

b) What are the test cases and passing criteria ?

No.	Test Case and Objective	Passing Criteria
1	App securely pairs select devices	Device is securely paired
2	App monitors network traffic by the device	Log displays important criteria (time, source, destination, protocol, packet length, packet info)
3	App monitors device vitals	Log displays device specific sensor and CPU data
4	App sends device vitals to the cloud	Cloud-based dashboard visualizes device data
5	ML model (on the cloud) fingerprints device features	Classification table of device features
6	ML model (at the edge) monitors device features	ML model deployed at the edge (user's device)
7	App detects anomalous behavior (e.g., MitM attack)	App identifies potential attack
8	App detects hardware malfunctioning (e.g., damaged sensors)	App identifies problematic sensors or other components

5. Project Timeline and Milestones

a) Propose the project schedule

No.	Scope of Work, Tasks, Tests, Specification and Deliverables	Months (T + N)	Status & Remarks
1	Requirements Gathering	+ 1	
2	UX /		
3	ML Model Development	+ 4	
Milestone 1: Completion of < Prototype Demo >		**T + 6**	
4	Full stack application	+ 1	
5	Model tuning and extensions	+ 3	
6	Integration with other infrastructure	+ 2	
Milestone 2: Completion of < Production Solution >		**T + 12**	
7	Extend development to industrial applications		
8			
9			
Milestone 3: Completion of < Industrial Prototype >		**T + 18**	
10	Complete development for industrial applications		
11			
12			
Milestone 4: Completion of < Scalable Product >		**T + 24**	

6. Project

Category	Description	Total Amount (S$)
A. Expenditure on in-company Manpower (EOM) (Will be reimbursed via validation of **salary** documents)	Senior Security Engineer ($120,000) Full Stack Engineer ($120,000) Data Analyst ($60,000)	$300,000
B. Expenditure on Third Party expenses such as purchases of Equipment/Software/hardware/Professional Services (Will be reimbursed via validation of **invoice** documents)	UX/UI product design ($60,000) Electronics assembly and configuration ($60,000)	$120,000
C. Other Operating Expenses (OOE)	Technology Management ($120,000) Hosting and licensing (TBD)	$120,000
	Total ed (A+B+C)	$540,000
(e.g., manpower, equipment) that the Applicant Company and/or collaborators (if any) will self-fund	-	-
	Total Self-Funding (D)	**-**
	Total Project Cost (A+B+C+D)	**$540,000**

7. Project Impact

a) What is the potential for the solution to be developed into a commercial product for industry-wide adoption?

The primary application proposes a project of 12 months, ideally from January 2022 to December 2022. The end solution will meet the requirements of the CSA's Challenge Statement for CS07: Unified IoT Security for Connected Products Utilising . It will be largely geared towards consumer electronics (i.e., smart homes). The project can be expanded further to industrial applications requiring a similar solution. This can be done in the subsequent 12 months (January 2023 to December 2023).

b) What is the estimated budgetary pricing of the solution after the project has been completed successfully?

No	Description of components	One-Time-Cost/ CAPEX (S$)	Annual Recurring Cost/OPEX (S$)	Price/Discount for current End User
1.	Software maintenance costs (TBD)			
2.				
3.				

c) What is the estimated international addressable market size for the solution?

Current estimated market size for AI applications in buildings and homes is $868M, growing at 22.6% CAGR in smart homes and 35.7% in commercial buildings segment (according to Frost & Sullivan)

8. Company Background

No	Category	Sub-category	Value
1	**Full company name**	Singapore Incorporated	Ambient Systems Pte. Ltd.
2	**Company global headquarter**	Country	Singapore
3	**Years of establishment**	Global	2020
		Singapore	2020
4	**Number of employees**	Global	4
		Singapore	4
5	**Core business activities**	Key products/services	Enterprise software, machine learning, IoT
		Existing clientele	2 LOIs with industrial facilities in Singapore

Shareholding Structure

No	Share Holder Name	Percentage of Holding	
1	Ivan Damnjanović	100%	Canadian
2			
3			
4			

9. Project Team

a) List all team members involved in this project

S/N	Name	Job scope in project (list at least 2-3 points)	in which Country	Company	Years of experience	Funding Needed (Y/N)
1	Ivan Damnjanović	• Technology management (product and IP) • System design and engineering • Team management	Canada / Singapore	Ambient Systems	4	Y
2		• Data analytics	Singapore / Singapore	Ambient Systems	2	Y
3	TBD	UX/UI product design	-	Ambient Systems	-	-
4	TBD	Senior Security Engineer	-	Ambient Systems	-	-
5	TBD	Full Stack Engineer	-	Ambient Systems	-	-
6	TBD	Electronics assembly	-	Ambient Systems	-	-

b) Does the core team reside in Singapore and project will be developed in Singapore? (Yes/No)

Yes

10. Others

a) Any other considerations that will bring merit to the proposed solution

Ambient Systems is a deep tech spin-off from National University of Singapore's (NUS) Graduate Research and Innovation Programme (GRIP).

We consider ourselves an enterprise software company that provides applied AI solutions for cyber-physical systems. We have patent-pending invention that leverages AI to overcome challenges in the built environment. These include AI-enabled planning, scheduling, and predictive maintenance, enhancing overall system performance, cybersecurity, and sustainability.

b) Does the proposed solution require longer term Research and Development (R&D)?

The proposal solution focused on the initial challenge statement, namely enhancing cybersecurity of smart home systems. Parallel to the original problem statement, we see an opportunity to develop similar technology for the industrial segments, particularly in AgTech, Healthcare, and Retail. We would be interested in pursuing R&D collaboration on developing similar solutions for these domains in parallel or after completion of the initial milestones.

Thank You